

```
mysql>
mysql> INSERT INTO Cancellations VALUES
-> (501,1002,'2026-09-11',4500.00);
Query OK, 1 row affected (0.01 sec)
```

```
mysql> SELECT
-> P.passenger_id,
-> P.name AS passenger_name,
-> P.email,
-> P.passport_no,
-> F.flight_id,
-> F.flight_number,
-> F.origin,
-> F.destination,
-> B.booking_id,
-> B.booking_date,
-> B.seat_number
-> FROM Passengers P
-> INNER JOIN Bookings B
-> ON P.passenger_id = B.passenger_id
-> INNER JOIN Flights F
-> ON B.flight_id = F.flight_id
-> WHERE F.origin = 'New York'
-> AND F.destination = 'London'
-> ORDER BY B.booking_date;
```

passenger_id	passenger_name	email	passport_no	flight_id	flight_number	origin	destination	booking_id	booking_date	seat_number
101	Rahul	rahul@gmail.com	P12345	1	AI101	New York	London	1001	2026-09-10	12A
102	Priya	priya@gmail.com	P67890	1	AI101	New York	London	1002	2026-09-10	12B
103	Sampath	sampath@gmail.com	P54321	3	AI303	New York	London	1003	2026-09-12	15C

```
mysql> CREATE VIEW vw_flight_occupancy AS
-> SELECT
->     flight_id,
->     flight_number,
->     origin,
->     destination,
->     total_seats,
->     available_seats,
->     ((total_seats - available_seats) * 100.0 / total_seats)
->     AS occupancy_percentage
-> FROM Flights;
```

Query OK, 0 rows affected (0.01 sec)

```
mysql> SELECT * FROM vw_flight_occupancy;
```

flight_id	flight_number	origin	destination	total_seats	available_seats	occupancy_percentage
1	AI101	New York	London	200	150	25.00000
2	AI202	Paris	Dubai	180	120	33.33333
3	AI303	New York	London	220	180	18.18182

3 rows in set (0.00 sec)

```

mysql> DELIMITER $$
mysql>
mysql> CREATE FUNCTION fn_check_seat_availability(
->     p_flight_id INT
-> )
-> RETURNS VARCHAR(20)
-> DETERMINISTIC
-> BEGIN
->     DECLARE seats INT;
->
->     SELECT available_seats
->     INTO seats
->     FROM Flights
->     WHERE flight_id = p_flight_id;
->
->     IF seats > 0 THEN
->         RETURN 'AVAILABLE';
->     ELSE
->         RETURN 'NOT AVAILABLE';
->     END IF;
-> END $$

```

Query OK, 0 rows affected (0.01 sec)

```

mysql>
mysql> DELIMITER ;
mysql> SELECT fn_check_seat_availability(1);

```

```

+-----+
| fn_check_seat_availability(1) |
+-----+
| AVAILABLE                     |
+-----+

```

1 row in set (0.01 sec)

```
mysql> DELIMITER $$
mysql>
mysql> CREATE TRIGGER trg_update_available_seats
-> AFTER INSERT ON Bookings
-> FOR EACH ROW
-> BEGIN
->     UPDATE Flights
->     SET available_seats = available_seats - 1
->     WHERE flight_id = NEW.flight_id;
-> END $$
```

Query OK, 0 rows affected (0.01 sec)

```
mysql>
mysql> DELIMITER ;
mysql> SELECT flight_id, available_seats
-> FROM Flights
-> WHERE flight_id = 1;
```

flight_id	available_seats
1	150

1 row in set (0.00 sec)

```
mysql> INSERT INTO Bookings
-> VALUES
-> (1004,1,103,'2026-09-15','18A');
```

Query OK, 1 row affected (0.01 sec)

```
mysql> SELECT flight_id, available_seats
-> FROM Flights
-> WHERE flight_id = 1;
```

flight_id	available_seats
1	149

1 row in set (0.00 sec)